

NOOR()

USER MANUAL

SongKeeper

The program is a Song management system.

It allows the user to:

- Add a song
- Search a song
- Delete a song
- Save songs into a file
- Load saved songs automatically.

At first, the user has a main menu.

```
~~~~~ SongKeeper Menu ~~~~~
1. Search a song.
2. Add a song.
3. Delete a song.
4. Save to file.
5. Exit.
Choose:
```

If invalid input is written:

```
~~~~~ SongKeeper Menu ~~~~~
1. Search a song.
2. Add a song.
3. Delete a song.
4. Save to file.
5. Exit.
Choose: n
Invalid option. Please enter a number from 1 to 5:
2

Enter title: |
```

At this point, the user has choices out of five options:

1. Search a song

If the user chooses this option, the user can search for songs using:

- Title
- Artist
- Album
- Genre
- Year

The program will display all matching songs with their information.

```
Song added.

~~~~~ SongKeeper Menu ~~~~~
1. Search a song.
2. Add a song.
3. Delete a song.
4. Save to file.
5. Exit.
Choose: 1

Search by:
1. Title.
2. Artist.
3. Album.
4. Genre.
5. Year.
6. Back.
Choose: 1

Enter search value: hello

Title: hello
Artist: adele
Album: 25
Genre: pop
Year: 2015
~~~~~

Title: hello
Artist: lionel richie
Album: can't slow down
Genre: soft rock
Year: 1984
~~~~~

~~~~~ SongKeeper Menu ~~~~~
1. Search a song.
2. Add a song.
3. Delete a song.
4. Save to file.
5. Exit.
Choose: 1

Search by:
1. Title.
2. Artist.
3. Album.
4. Genre.
5. Year.
6. Back.
Choose: 4

Enter search value: r&b

Title: blinding lights
Artist: the weekend
Album: after hours
Genre: R&B
Year: 2019
~~~~~
```

The search is case-insensitive, meaning capital or lower letters does not matter. If no song is found, an error message will appear.

```
~~~~~ SongKeeper Menu ~~~~~
1. Search a song.
2. Add a song.
3. Delete a song.
4. Save to file.
5. Exit.
Choose: 1

Search by:
1. Title.
2. Artist.
3. Album.
4. Genre.
5. Year.
6. Back.
Choose: 1

Enter search value: hello

Error: No song in the collection, please add a song first.
```

If no matching is found:

```
Enter search value: Blinding Lights  
  
No matching song found.
```

If invalid option is written:

```
~~~~~ SongKeeper Menu ~~~~~  
1. Search a song.  
2. Add a song.  
3. Delete a song.  
4. Save to file.  
5. Exit.  
Choose: 1  
  
Search by:  
1. Title.  
2. Artist.  
3. Album.  
4. Genre.  
5. Year.  
6. Back.  
Choose: n  
Invalid option. Please enter a number from 1 to 6:  
█
```

2. Add a song

If the user chooses this option, the user will be asked to enter:

- Title
- Artist
- Album
- Genre
- Year

The program will create a new song and store it in the collection.

```
~~~~~ SongKeeper Menu ~~~~~  
1. Search a song.  
2. Add a song.  
3. Delete a song.  
4. Save to file.  
5. Exit.  
Choose: 2  
  
Enter title: blinding lights  
Enter artist: the weekend  
Enter album: after hours  
Enter genre: R&B  
Enter year: 2019  
  
Song added.
```

The limits of the inserted data are:

- Year cannot be negative
- Duplicated songs are not allowed
- Songs are checked using title and artist.

If duplicate song exists, the program will display an error message.

```
~~~~~ SongKeeper Menu ~~~~~
1. Search a song.
2. Add a song.
3. Delete a song.
4. Save to file.
5. Exit.
Choose: 2

Enter title: Hello
Enter artist: Adele
Enter album: 25
Enter genre: soul
Enter year: 2015

Error: Song already exists.
```

If invalid year input, the program will display an error message and to enter a (positive) number.

```
2

Enter title: Hello
Enter artist: Adele
Enter album: 25
Enter genre: pop
Enter year: n
Invalid year. Please enter a (positive) number:
2015

Song added.
```

3. Delete a song

The user will be asked to enter the title of the song to delete.
If only one matching song exists, it will be deleted immediately.

```
~~~~~ SongKeeper Menu ~~~~~
1. Search a song.
2. Add a song.
3. Delete a song.
4. Save to file.
5. Exit.
Choose: 3

Enter title of song to delete: adel

1. Adel - adele
Song deleted.
songs.txt file is updated.
```

If multiple songs with the same title exists, the program will display all matching songs and ask the user to choose which song should be deleted. Or there is another option to go back to the main menu if the user doesn't want to delete anything.

```
~~~~~ SongKeeper Menu ~~~~~
1. Search a song.
2. Add a song.
3. Delete a song.
4. Save to file.
5. Exit.
Choose: 3

Enter title of song to delete: hello

1. hello - adele
2. hello - lionel richie
0. Back
Choose song number to delete: 0
Delete cancelled.
songs.txt file is updated.

~~~~~ SongKeeper Menu ~~~~~
1. Search a song.
2. Add a song.
3. Delete a song.
4. Save to file.
5. Exit.
Choose: ☐
```

```
~~~~~ SongKeeper Menu ~~~~~
1. Search a song.
2. Add a song.
3. Delete a song.
4. Save to file.
5. Exit.
Choose: 3

Enter title of song to delete: hello

1. hello - adele
2. hello - lionel richie
0. Back
Choose song number to delete: 2
Song deleted.
songs.txt file is updated.
```

After deletion, the song will be removed from memory and the updated collection is saved automatically into "songs.txt".

If no matching song exists, an error message will appear.

```
~~~~~ SongKeeper Menu ~~~~~
1. Search a song.
2. Add a song.
3. Delete a song.
4. Save to file.
5. Exit.
Choose: 3

Enter title of song to delete: adel

Error: Song was not found.
```

4. Save to file

If the user chooses this option, all songs are saved into “songs.txt”.

```
~~~~~ SongKeeper Menu ~~~~~
1. Search a song.
2. Add a song.
3. Delete a song.
4. Save to file.
5. Exit.
Choose: 4

songs.txt file is updated.
```

Each song is stored in one line using semicolon-separated values.

```
hello;adele;25;pop;2015
blinding lights ;the weekend;after hours;R&B;2019
hello;lionel richie;can't slow down;soft rock;1984
```

The file allows songs to remain saved even after the program closes.

5. Exit

If the user chooses this option, all songs are automatically saved, the program will terminate, and the user exits the program.

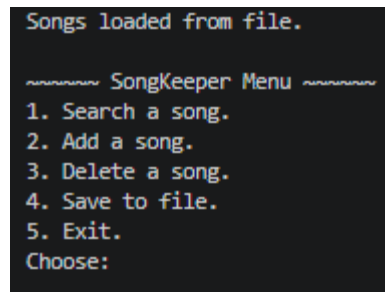
```
~~~~~ SongKeeper Menu ~~~~~
1. Search a song.
2. Add a song.
3. Delete a song.
4. Save to file.
5. Exit.
Choose: 5

songs.txt file is updated.
Program stopped.
```

Automatic File Loading:

When the program starts, it automatically tries to load songs from “songs.txt”.

If the file exists, all previously saved songs are stored automatically.



```
Songs loaded from file.

~~~~~ SongKeeper Menu ~~~~~
1. Search a song.
2. Add a song.
3. Delete a song.
4. Save to file.
5. Exit.
Choose:
```

If the file does not exist yet, the program continues normally.

Error Handling:

The program uses exception handling to prevent crashes.

Example of handled errors:

- Duplicated songs
- Invalid year value
- Deleting non-existing songs
- File opening errors

Error messages are displayed directly to the user, as seen above in the pictures.

Program Execution:

The project is compiled using:

```
g++ main.cpp SongKeeper.cpp -o SongKeeper.exe
```

The program is started using:

```
.\SongKeeper.exe
```

The user communicates with the program through the terminal window using keyboard input.